BATTERY SELECTION · NINE MODELS

# Better forecasts did not reliably improve choices.

Early cycles in. Later measurements hidden.

Could the models choose which sample retained more capacity?

**15/16**

## Lower forecast error

than the carry-forward baseline



**3/16**

## More correct choices

than the baseline's 5 / 12 pair choices



**11 / 16 also beat the stronger trend baseline on forecast error.**

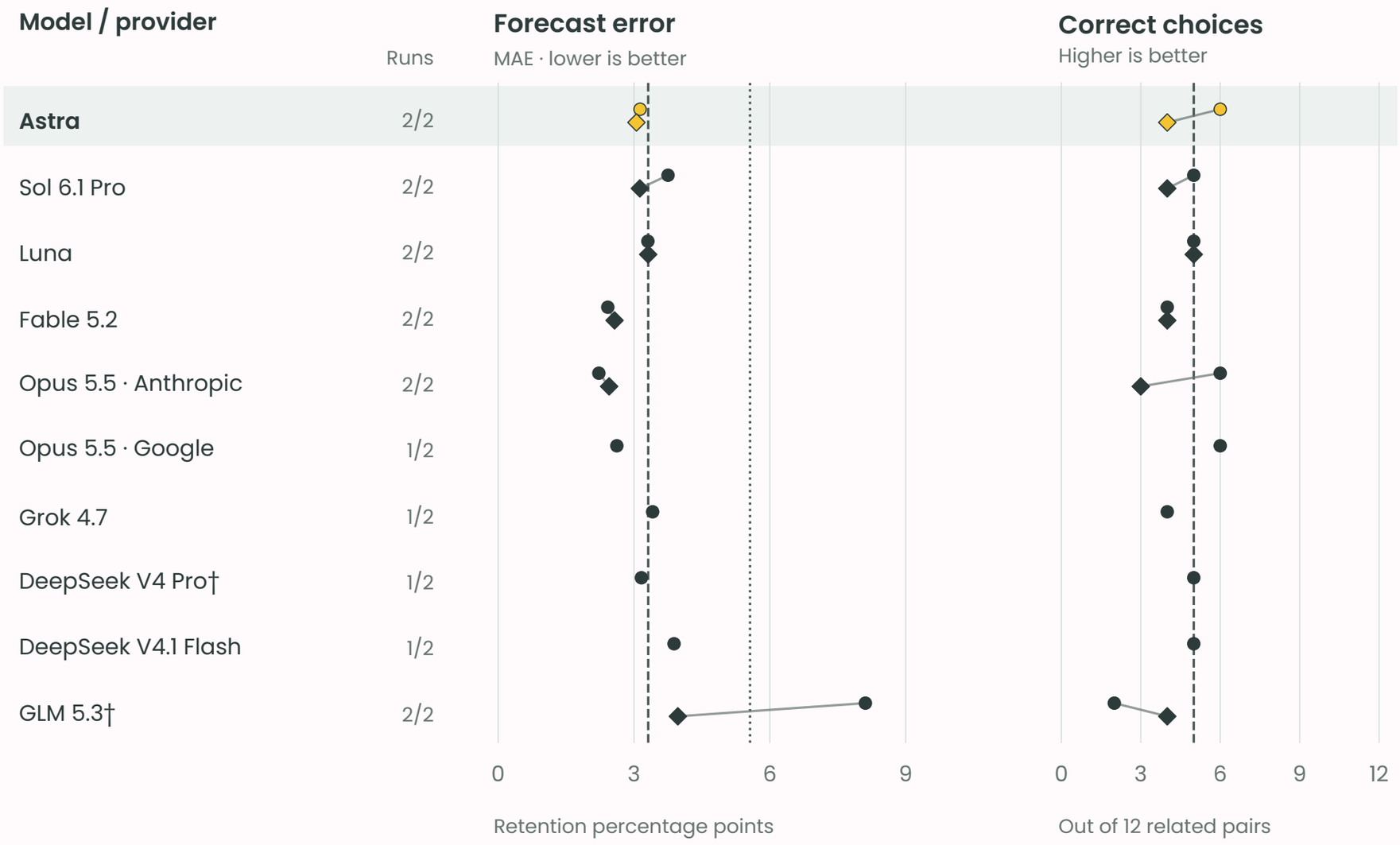
16 completed attempts on the same 10 traces and 12 related pairs.

Counts describe attempts, not independent experiments. Unequal repeats.

SAME INPUTS · HIDDEN OUTCOMES

# Battery: forecasting vs choosing

Cycles 1–50 to retention at cycles 351–360




**Baselines: trend 3.31 pp; carry-forward 5.56 pp. Both choose 5 / 12.**

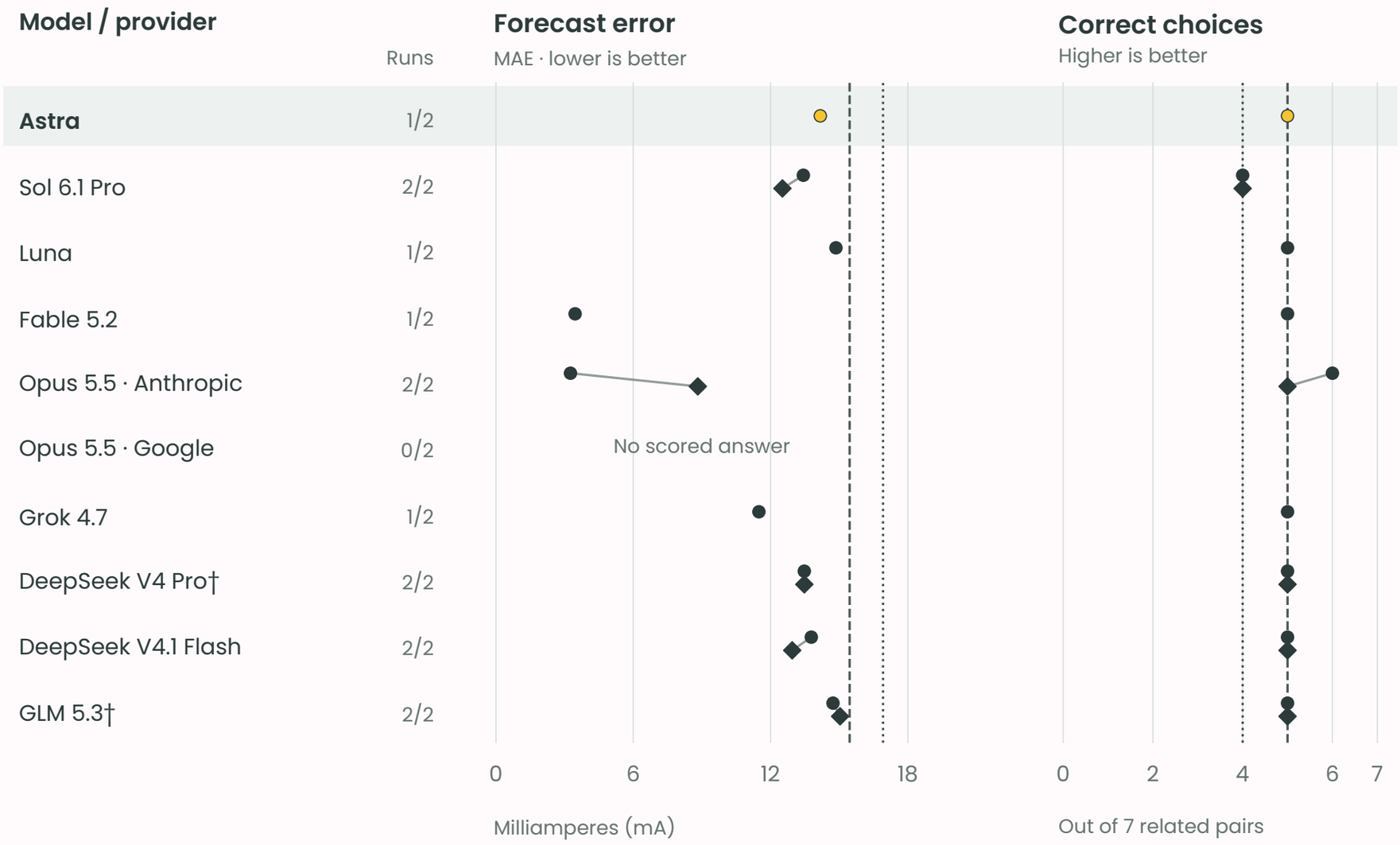
Circles: attempt 1 · Diamonds: attempt 2 · Dashed: trend / linear tail · Dotted: carry-forward

† Format adaptation. Unequal completions; rows are not a ranking. Full settings in repository.

SAME INPUTS · HIDDEN OUTCOMES

# Catalyst: forecasting vs choosing

Partial scan through 0.600 V to signed current at 0.800 V




**Baselines: linear tail 15.45 mA, 5 / 7; carry-forward 16.91 mA, 4 / 7.**

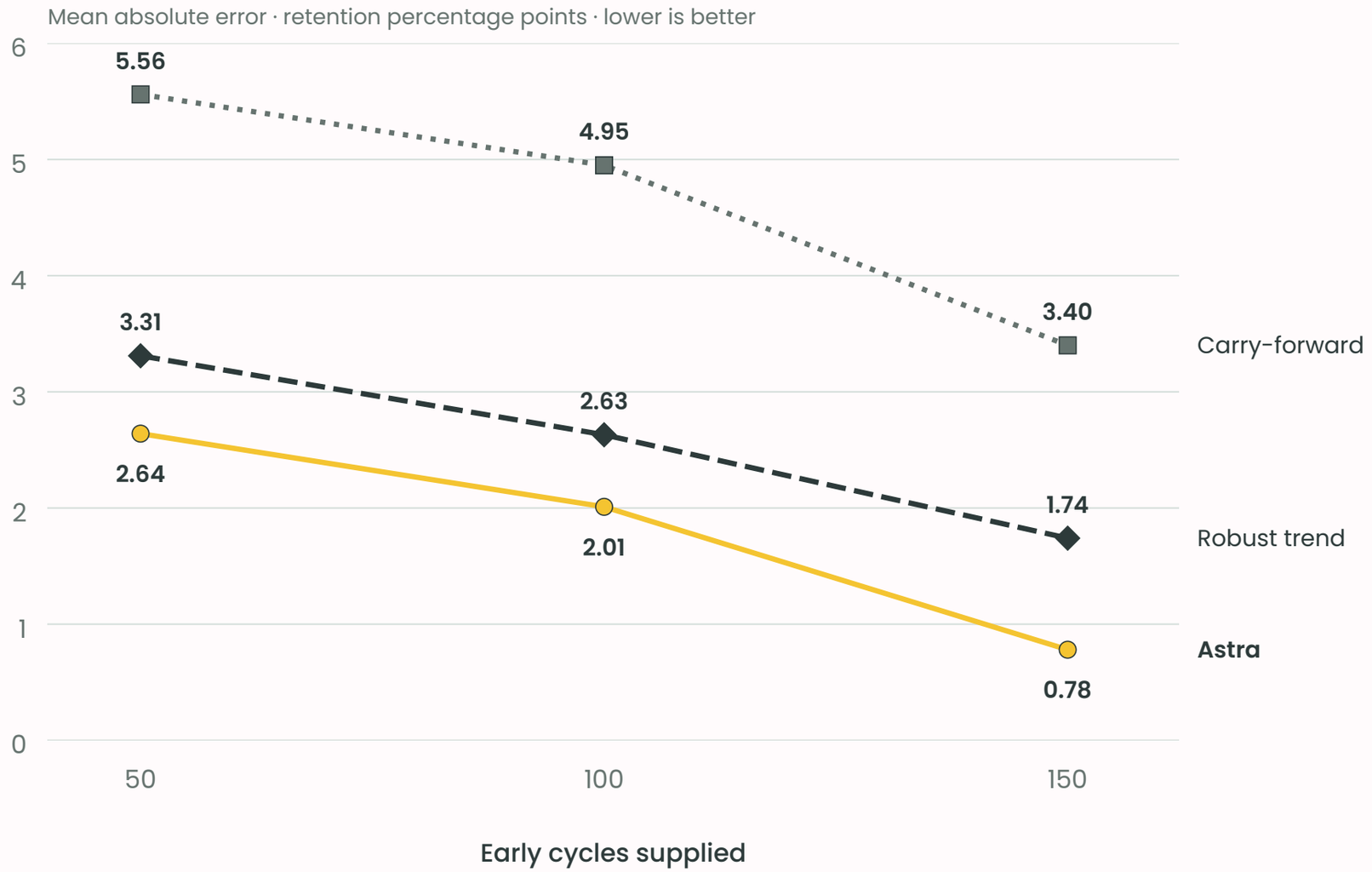
Circles: attempt 1 · Diamonds: attempt 2 · Dashed: trend / linear tail · Dotted: carry-forward

† Format adaptation. Unequal completions; rows are not a ranking. Full settings in repository.

ASTRA FOLLOW-UP · SEPARATE SUBSCRIPTION STUDY

# More early data improved Astra’s battery forecasts.

Same 10 traces. Later measurements stayed hidden.




Separate runs on reused records; do not pool with the multi-model comparison.

Mean-window target shown. A median-window check changes the ordering at 150 cycles.

ASTRA FOLLOW-UP · CELL SELECTION

# Astra reached 6/6 on cell selection.

Find the cell with the lowest later 0.5C grading capacity.

Calibration examples supplied; target-cell outcomes hidden.




**Correct choices across six attempts on three reused cell groups.**

Prompt, methods and calibration validation changed together after outcomes were known.

Same 8 cells; a development follow-up, not a controlled causal test or training intervention.